

Limit vs. Market Order Trading on the Saudi Stock Market

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Comments are welcomed.

Market vs. Limit Order Trading on the Saudi Stock Market

Abstract

This paper uses an order data set to analyze trading by limit versus market orders on the Saudi Stock Market. Building on the literature on a trader's choice problem in order driven markets, we interpret the data using a *Random Utility Model* and approach the problem statistically using a logit model. The performance of the orders resulting from the decisions of traders also is examined. The empirical results show that the probability of placing a market order is negatively related to the inside spread and order size, and is positively related to order imbalance and previous same-side market orders. The performance measures indicate that limit order strategies perform better than a market order strategy.

Keywords: Market Microstructure, Order-driven Market, Order Placement, Order Performance.

JEL classification: G15

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1. Introduction

In a pure order driven market, the investor must decide between placing a limit order or trading via a market order. Market orders are executed with certainty, but pay an implicit price for immediacy. Limit orders can profit from supplying liquidity, but bear the risk of adverse selection, free option and non-execution. The choice between limit and market orders depends on market conditions and the propensity of the investor to trade (his relative patience).

The objective of this paper is to analyze trading by limit versus market orders on the Saudi Stock Market (SSM). The SSM is a pure order driven market with no central exchange, regulated brokers or designated market makers. Traders who voluntarily provide liquidity to the market by posting limit orders rather than market orders sustain the market. Our data set allows us to identify market and limit orders, and other characteristics that are related to traders and market conditions. We interpret the data using a *Random Utility Model*, and empirically approach the problem facing the trader in this market using a logit model. We also examine how the orders resulting from the decisions of traders perform in this market by using different performance measures. This paper uses the same data set used by Al-Suhaibani and Kryzanowski (2000a,b). Al-Suhaibani and Kryzanowski (2000a) study the interaction between the order book and order flow, and limit order execution on the SSM. Al-Suhaibani and Kryzanowski

(2000b) examine the dynamic behavior of orders and stock prices in order to assess the information content of newly submitted orders on the SSM. In contrast, this paper focuses on the choice of order type (limit or market) by a trader and the resultant profitability of such choices.

Our analysis builds on several theoretical and empirical studies on the investor's choice between market and limit orders in a pure order-driven market. Cohen, Maier, Schwartz and Whitcomb (1981) model the investor order placement strategy in a market with no asymmetric information. Angel (1995) models an informed trader's choice between limit and market orders to trade a fixed number of shares. Harris (1998) solves optimal dynamic order submission strategies for trading problems faced by three stylized traders; namely, an uninformed liquidity trader, an informed trader, and a value-motivated trader. Foucault (1999) derives a game theoretic model of order placement and price formation, given no asymmetric information, in a dynamic order driven market. Parlour (1998) characterizes dynamic equilibrium in a market where traders optimally choose the type of order to submit given no asymmetric information, and the stock's underlying value is fixed. Handa, Schwartz and Tiwari (1997) explicitly model the choice of trading strategy (market vs. limit orders) faced by an uninformed trader in a pure order driven market with asymmetric information, and study its impact on price formation.

In the empirical literature, Biais, Hillion and Spatt (1995) examine order flow dynamics on the Paris Bourse. Hollifield, Miller, and Sandås (1996) empirically analyze a generalization of Foucault's model that allows orders to last over multiple periods. Lo, MacKinlay and Zhang (1998) develop several econometric models of limit order execution times using survival analysis. Recently, Ahn, Bae, and Chan (1999) provide

results on the relationship between transitory volatility and market depth and order flow composition in the Hong Kong stock market.

The remainder of this paper is organized as follows. Section 2 describes the market and data set. We perform the empirical analysis and report the findings in sections 3 and 4. Section 5 concludes the paper.

2. Market and Data Description

2.1 The Structure of the Market

The SSM is relatively new compared to the stock markets in the developed countries. The market started with one company in 1952, and trading was completely unregulated until 1984. The regulations transferred share trading from the unofficial brokers to the banks under the supervision of SAMA (Saudi Arabian Monetary Agency), the central bank in Saudi Arabia. The second major development in trading on the SSM post-market-regulation was the establishment of an electronic trading system known as ESIS (Electronic Security Information System). After the startup of ESIS in August 1990, the banks established twelve Central Trading Units (CTUs). All the CTUs are connected to the central system at SAMA.

Trading lasts for four hours per day, divided into two daily sessions for Saturday through Wednesday, and for one two-hour session on Thursday. Sell and buy orders are generated from the incoming sell and buy order packages. An order package can have one or many firm orders, each differentiated by parameters such as quantity, price and validity period. Order packages entered into the system may be valid for a period from 1 to 12 days.

The market opens with a *call market*. A continuous auction market, where market orders by public investors are transacted with the limit orders of other public investors, follows this.¹ For limit orders, price-then-time priorities rules are strictly enforced. An investor can adjust order prices and their quantities. With each change, the order loses its time priority. Aggressive sell (buy) orders can walk down (up) the limit order book. When an order is partially executed, any unexecuted balance is automatically placed in a new order at the same price and with the same execution priority as the original order. The order package can be executed fully or partially through more than one transaction at different times, with different orders, and maybe, with different prices.

The electronic limit order book is not fully visible to investors. They can view only the best two bids and asks in an aggregate format (i.e. only the best quote with all quantities available at that quote). They also can view the price, quantity, and time of last trade. The status of the best quotes and quantities is updated (almost instantaneously) on bank screens each time an order arrives, is canceled, or is executed. The terminals and big screens where traders can monitor the market are only available in the CTUs and other designed branches in various parts of the country that are connected to the CTU. This level of transparency on the SSM hides all firm orders outside the two best quotes. For example, a firm order to buy with a price less than the second best bid is hidden but

¹ The standard market order (order to buy or sell a given quantity for immediate execution at the current market price, without specifying it) is not accepted by the system at the SSM. To obtain immediate execution, traders in this market use marketable limit orders. A marketable limit order is a limit order with a limit price at or better than the prevailing counterparty quotes. For a marketable buy (sell) order, the price must equal or better the current ask (bid). Since marketable and market orders are essentially the same, we use the term market orders when referring to marketable orders in the remainder of the paper.

becomes visible when all the quantity at the first best quote is executed. The order can also be executed while it is hidden by an aggressive market sell order.²

The tick size for all stocks in the market is SR 1 ($\approx$ 27 cents). Transaction fees, which are charged on each side of the trade, have a minimum of SR 25 ($\approx$ \$ 6.66), and range between 0.5% and 0.1% of trade value. During continuous trading periods, firm orders must be priced within $\pm$ 10% of the opening price of the given trading period. If no opening price exists for that period, the opening price defaults to the previous day's closing price.

Only the CTU employees have the right to enter orders directly into the system. Investors in the SSM consist of public investors and bank phone customers.³ The latter group of customers have an agreement with the banks to change the price and firm quantity of their submitted orders at any time simply by calling their Bank's CTU. As a result, they are less affected than other public traders by the free trading option associated with limit orders since they can change the condition of their orders very quickly before they are "picked off" when new public information arrives. This type of trader includes value motivated investors (e.g. mutual funds), informed traders, and many technical traders.

Following the second trading session, transactions are routed for settlement. The settlement date depends on the type of ownership document. *Ishaar*, which can be retained in the system for future sale or printed and given to the investor, are delivered

² Unlike some trading systems, ESIS does not allow traders to intentionally hide orders that are part of the best two quotes.

³ SAMA does not allow banks to grant their customers access to the system via any computer network.

next day morning, while certificates take from two days to one week or more to be delivered. *Ishaar* takes less time because it can be handled electronically through ESIS Fully Automated Share Transfer (ESISFAST), while the new certificates have to be issued from the company's share registration department.⁴

2.2 The Data Set

The data set provided by the SAMA includes all submitted market and limit orders for 56 stocks during the period from 31 October 1996 to 14 January 1997. For each order, the data set reports security code, the date and time of creation, sell-buy indicator, limit price, quantity, and date and time when the order was terminated. The data set also identifies the order package that generates the order. We use the date and time of termination, and the price and quantity of orders to identify orders that are executed. The identified transaction records perfectly match the published daily transaction statistics. Given the order data set, the state of the book can be determined at any point of time during the sample period.

For the executed orders, we are able to differentiate between the market and (executed) limit orders using the characteristics of the order and the order book. A market order typically has a zero duration, can only be placed during the continuous trading session, and has a price at or better than the prevailing opposite quote. We double-check our identification procedure by examining the change in the state of the book at the time of order submission. Market orders clear the book, whereas limit orders add to the book. Table 1 presents summary statistics for the data set.

⁴ More on the microstructure of the SSM is found in Al-Suhaibani and Kryzanowski (2000a).

Please place Table 1 about here.

Most of the analysis in this paper requires orders to be preceded by a valid bid-ask spread. A valid bid-ask spread is defined herein as one in which both the bid and ask prices are established and are visible to market participants. Only 5.67% of the orders entered during the continuous trading session fail to meet this criterion. The summary statistics for the restricted data set (not reported herein) are similar to the statistics reported in Table 1.

Table 1 reveals that buyers use market orders more often than sellers. This could be attributed to the rising market during the sample period. The all-market index rose by 9.23% over the studied time period. More than half the limit orders do not execute. Size statistics show that the largest percent of market and limit orders have a size of between 100 and 500 shares. A large percentage of the orders are parts of packages. If the number of orders per package is interpreted as an indicator of trader activity, then the data suggests that less active traders usually use limit orders, and more active traders typically use market orders.

3. Logit Analysis of Choice of Type of Order

In our pure order-driven market, the trader faces a tradeoff. If he places a market order, execution is assured at the prevailing opposite quote, but the trader pays an implicit price for immediacy. If the trader places a limit order, there are different possible scenarios. Consider a limit order to buy.⁵ If the order executes during the trading window due to arrival of a liquidity-motivated seller, the limit order trader benefits because he

⁵ The scenarios are symmetrical for a limit order to sell.

saves paying the spread. However, if the limit order executes due to the arrival of an informed trader with a valuation below the limit price, the limit order trader loses. A similar situation occurs if the limit order becomes mispriced because public information causes the stock value to fall below the limit price. This risk might exist even if all traders have the same information at a given point of time. Moreover, the limit order may not execute during the trading window. This may occur because the limit price position is far below the market, the stock value rises, or because no liquidity-motivated seller arrives during the trading window. In all cases, a decision has to be made at the end of the trading window on whether to trade at the prevailing ask or to forego trading. This decision has cost implications that depend on the relative patience of the trader. The more patient trader will be more willing to forego trading, and hence bear less non-execution cost.

Given these possibilities, a trader chooses to trade using a limit order rather than a market order if his expected utility from placing the limit order, U_L , exceeds that from placing a market order, U_M . The expected utilities depend critically on the probabilities of his order executing against an informed or liquidity trader. In our data set we only observe the trader's choice. An indicator, y , equals one if the order is a market order and zero if it is a limit order. The observed choice reveals which one provides the greater utility, but not the unobservable utilities. We also observe other variables that theory suggests are likely to affect this decision. In particular, we observe variables related to market conditions before order placement (such as spread and depth) and some order characteristics (such as order size and direction).

3.1 The Model for Trader Choice of Order Type

The data on the trader's choice can be interpreted using the *Random Utility Model*.⁶ In this model, the utility of the trader is given by,

$$U_i = V_i(x) + \varepsilon_i \quad i = M, L \quad (1)$$

where x is a set of explanatory variables that are used to explain the occurrence of market and limit orders and ε_i is the error term. The function defined in $V_i(x)$ is commonly assumed to be linear, that is, $V_i(x) = x\beta_i$. This formulation implies that:

$$\begin{aligned} \text{Prob}[y = 1 | x] &= \text{Prob}[U_M > U_L | x] \\ &= \text{Prob}[x(\beta_M - \beta_L) + \varepsilon_M - \varepsilon_L > 0 | x] \\ &= \text{Prob}[x\beta + \varepsilon > 0 | x] \end{aligned}$$

Assuming that ε has a standard (symmetric) logistic distribution with mean zero and variance one, we get the logit model:⁷

$$\begin{aligned} \text{Prob}[y = 1 | x] &= \text{Prob}[\varepsilon > -x\beta] = \text{Prob}[\varepsilon < x\beta] \\ &= \Lambda(x\beta) \end{aligned} \quad (2)$$

where $\Lambda(x\beta)$ is the logistic cumulative distribution function, $e^{x\beta} / (1 + e^{x\beta})$. The maximum likelihood method is used to estimate the parameter vector, β . In interpreting the estimated model, the marginal effect of x on the probability is given by $\Lambda(x\beta)[1 - \Lambda(x\beta)]\beta$.

In our empirical analysis, we test several theoretical predictions concerning the relationship between the decision to trade by market or limit orders and a set of

⁶ Maddala (1983) and Greene (1997).

explanatory variables. The specification of the set of explanatory variables, x , includes a constant and six variables. The first four variables reflect the market conditions prevailing prior to submission of the order, and the last two variables are related to the trader himself. Table 2 presents summary statistics for these six explanatory variables. We now discuss each of these variables in turn.

Please place Table 2 about here.

A. *Inside spread*:

The theoretical motivation for including the spread is obvious. Although the theoretical models referred to in the introduction differ in their assumptions regarding types of traders, trading mechanisms, preferences, information structures and dynamics, all of the models predict that traders in a pure order driven market are more likely to submit market (limit) orders if the spread is tight (wide). As Cohen *et al* (1981) and Handa *et al* (1997) explain, when the spread becomes smaller the “*gravitational pull*” exerted by the existing opposite quote dominates and makes trading via a market order more attractive. Handa and Schwartz (1996) show that smaller inside spreads reduce short-run volatility and hence increase the tendency to trade via market orders. Parlour (1998) demonstrates that wider spreads enlarge the feasible set of limit order prices. Given price and time priority rules, traders having a larger opportunity set can increase the probability of executing their limit orders simply by undercutting or overbidding the prevailing quotes. We compute the *inside spread* as $A - B$, where A and B are the prevailing first best ask and bid at time of order entry, respectively.

⁷ Assuming that ε_i has a normal distribution with mean zero and variance one produces a probit model.

B. *Order imbalance*:

Order imbalance can affect the probability of executing the limit order, and hence the trader's decision. Handa *et al* (1997) show that as the proportion of buyers in the market becomes smaller, the non-execution risk for the buyer is lower. They also show that the relative attractiveness of trading via a market order at the posted ask, or via an aggressive limit order increases as this proportion increases. Foucault (1999) also reaches the same prediction. Similarly, in Parlour (1998), the trader's decision to submit a market or limit order depends critically on the thickness of both sides of the book at the time of order placement. Any event that changes the balance of book depth alters the probability of execution of the subsequent order, and therefore the decision of the next trader. *Order imbalance* is calculated herein as k for a buy order and $(1-k)$ for a sell order, where $k = Q_b / (Q_b + Q_s)$, and Q_s (Q_b) is the number of shares offered at A (demanded at B) at time of order entry. Thus, as k increases, the probability of executing a limit buy order decreases, and a market buy order becomes more attractive.

C. *Volatility*:

We distinguish between transitory and fundamental volatility. Transitory volatility is related to trading activities and is mean-reverting. Uninformed impatient traders can cause prices to bounce between the bids and asks. Fundamental volatility is related to changes in the stock's fundamental value which have permanent effects on prices. We are interested in testing the theoretical prediction of Handa and Schwartz (1996) regarding the relation between short-term price volatility and the placement of limit orders. Transitory volatility encourages more investors to submit limit orders rather than market orders. However, higher fundamental volatility, according to Harris (1998)

and Handa *et al* (1997), makes limit orders less attractive since volatility increases limit order option values.⁸ Hence, the length of the time interval used to measure volatility should be short enough for the volatility measure to differentiate transitory volatility from permanent volatility. Following Lo *et al* (1997), we use the ratio of trades (market orders) in the last half-hour to trades in the last one-hour as a proxy for *transitory volatility*.

D. *Prior market order*:

We include a *prior market order* variable because the history of trade pattern may provide information to market participants and may affect their decisions. The variable is computed as the ratio of the market orders that were initiated by the same side of the market to the total market orders submitted during the last half-hour of trading. The motivation for including this variable is the predicted positive correlation between past and current market orders. The correlation could arise from different effects. Biais *et al.* (1995) put forward three explanations for such observed behavior. First, the succession of identical types of orders could reflect strategic order splitting, either to reduce the market impact of a non-informational trade, or to get the most benefit from private information about the value of the stock. Second, if different traders are imitating each other, the cause of the correlation is the order flow itself. Finally, traders could react similarly to the same events related to a particular stock or the economy as whole. The Parlour queuing model (1998), which assumes no asymmetric information, provides a different explanation. A market buy order reduces the depth at the ask and increases the

⁸ In contrast, Foucault (1999) predicts that limit orders are positively related to asset volatility. He shows that high volatility increases the limit order option value, and hence the spread. However, at a wider spread, more traders find it optimal to trade using limit orders.

probability of executing a limit sell order next. As a result, the next trader, if she is a seller, is more inclined to submit a limit sell order.

E. *Order size:*

The variables that are specific to the trader are those that affect his propensity to trade (or his relative patience). Harris (1998) shows that a trader will be less patient if he faces a deadline, has material private information, is risk averse, competes with others or values his time. Unfortunately, we do not observe variables that can be used as measures for these factors. One variable that is related to the relative patience of the trader is order size. Larger order size (Number of shares in the order) may require the trader to be more patient because such an order will be more difficult to execute. The pure order driven market models do not have predictions regarding order size since they usually assume a fixed number of shares. However, the empirical evidence by Harris and Hasbrouck (1996) and Hollifield *et al* (1996) finds that limit orders tend to be larger.

F. *Orders per package:*

The orders per package variable is defined as the number of orders in the package that generated the submitted order. This variable is used to proxy for the degree of trader activity in the market, where large numbers indicate more active traders. More active traders tend to monitor the market more closely and are expected, as in Foucault, Röell and Sandås (1999), to seize opportunities using market orders. However, the more active traders could be traders who seek to profit from market making by placing limit orders and monitoring the market to minimize the free option risk.

3.2 Empirical Findings for Trader Choice of Order Type

We estimate the logit model using the pooled data for all of the 56 stocks in the sample. The logit regression coefficients and marginal effects are reported in Table 3, and the predicted probabilities are plotted in Figure 1.

Please place Table 3 and Figure 1 about here.

Consistent with the predictions of the theoretical models, the logit regression results show a negative relationship between the probability of placing a market order and the spread. At the sample means of the inside spread of 1.914 and 1.856, the effect on the probability is -0.0325 and -0.0155 for buy and sell orders, respectively. Figure 1 clearly shows that the probability of placing a market order decreases as inside spread increases from one to four.

Regarding order imbalance, the logit regression results show that when the order imbalance increases in favor of the other side of the market, traders are more likely to submit market orders. When the order imbalance is at its sample mean of 0.522 and 0.503, the effect on the probability of placing market orders is positive and is equal to 0.106 and 0.1336 for buy and sell orders, respectively. Figure 1 shows a pattern of increasing probabilities of placing a market order as the order imbalance increases from 0 to 1. The results on inside spread and order imbalance are consistent with the empirical findings of Biais *et al* (1995) who find “the conditional probability that investors place limit orders (rather than hitting the quotes) is larger when the bid-ask spread is larger or the order book is thin”.

The effect of transitory volatility is the only asymmetric effect. Table 3 illustrates that, apart from volatility, all explanatory variables have matching directional effects on the probability of placing buy and sell market orders. The marginal effect of volatility on the probability is positive for buy orders and negative for sell orders. It is apparent from Figure 1 that buyers (sellers) are more likely to submit market (limit) orders as volatility increases. These estimates imply that sellers in our market behave as Handa and Schwartz (1996) predict while buyers do not. However, this asymmetric effect can be explained using their analytical framework. Rising markets increase (decrease) the probability of execution and decrease (increase) the nonexecution cost for limit sell (buy) orders, and vice versa. Therefore, when traders expect upward or downward movements in prices, they behave differently. We are akin to this explanation due to the optimistic expectations of traders during the sample period.⁹ This optimism causes traders to expect an upward movement in prices, which makes limit sell orders less desirable and market buy order more desirable as short-run volatility increases. The result is consistent with the findings of Ahn *et al* (1999) that traders tend to place market buy (sell) orders when prices move up (down).

The prior market order variable has the strongest marginal effect on the probability of placing a market order. At the sample means of 0.45 and 0.452, the marginal effect of prior market order on the probability of placing a market order is positive and equal to 0.373 and 0.3134 for buy and sell orders, respectively. Figure 1

⁹ Oil prices and government fiscal policies are two major factors affecting economic activity in the country. The government budget (the most important financial event in the country) is usually announced on the first of January. Due to high oil prices in 1996, traders expected good news in the budget announcement.

clearly illustrates this positive effect as the value of the prior market order variable increases from 0 to 1. This result is consistent with the results of Biais *et al* (1995) who find that trades on one side of the market are most frequent after trades on the same side of the market. It also is consistent with the results of Ahn *et al* (1999) who find that order flow composition (market or limit) of buy and sell orders are positively autocorrelated.

Table 3 and Figure 1 show a marginally negative effect for order size, and a marginally positive effect for the number of orders per package on the probability of placing a market order. Execution difficulties for large orders require traders to be more patient and may explain the size effect. Given market thinness, large market orders can only be partially executed and only with high market impacts. Hence, to execute large orders with the lowest market impact, traders are more likely to use limit orders. This result is consistent with that of Harris and Hasbrouck (1996), who find limit orders tend to be larger than market orders. On the contrary, more active traders watch the market more closely and are expected to seize opportunities using market orders.

4. Performance of Limit versus Market Orders

After analyzing the trader's decision to trade using a market or limit order, it is natural to proceed to examine the performance of orders resulting from this decision. As discussed earlier in section 3, the limit order trader gains from liquidity events and suffers from information events that cause limit order execution. Furthermore, the non-execution of limit orders has a cost that depends on the relative patience of the trader. More patient traders have lower non-execution costs. Hence, for given liquidity-driven price changes,

Indeed, the SSM rose by 27.8 percent in 1997, its second best performance since it was regulated in 1984. The NCFEI Index rose to 195.89 points by year-end 1997, from 153.10 points at the end of 1996.

patient traders would prefer to trade via limit orders and relatively impatient traders would prefer to trade via market orders.

In this section, we examine the performance of executed and non-executed limit orders relative to market orders. The purpose is to determine whether the losses from information events dominate the gains from liquidity events, to find the size of non-execution costs in this market, and to determine whether a limit order strategy can outperform a market order strategy.

4.1 The Relative Performance Measures

Our analysis use measures employed by two earlier studies on order performance. Handa and Schwartz (1996) conduct an experiment where they replay the transaction record of thirty Dow Jones Industrial firms to assess the profitability of entering experimental, one-share market and limit orders. Their analysis is based on the difference between limit order return, δ_l^1 , and market order return, δ_m^1 , over three-day investment windows. The return is defined as

$$\delta_i^1 = \log(p_{e+3}) - \log(p_e) \quad i = m, l \quad (3)$$

where p_{e+3} is the opening price on Day 3 after execution, and p_e is the execution price for the order. Market orders are executed at the time of order submission, while limit orders can be executed during the trading window. For unexecuted orders, the execution price does not exist. These orders embody the real costs of a foregone trade. To avoid the selection bias that would result from dropping unexecuted limit orders from the sample, their measure embodies a strategy in which canceled or expired orders are resubmitted as a market order at the time of cancellation or expiration.

Using an actual sample of NYSE orders, Harris and Hasbrouck (1996) analyze the performance of market and limit orders using a similar measure that compares the actual or imputed execution price with the opposite-side quote at the time of order placement. It is defined as:

$$\delta_i^2 = \begin{cases} A_o - P_e & \text{for a buy order} \\ P_e - B_o & \text{for a sell order} \end{cases} \quad i = m, l \quad (4)$$

where B_o and A_o are the bid and ask quotes prevailing at the time of order submission.

This measure also follows the above approach for non-executed orders. This ex ante measure is designed to estimate the prospective benefit of using a limit order versus a market order that immediately executes against the prevailing opposing quote. The measure will be zero for market orders, $\delta_m^2 = 0$, because a market order cannot better the opposite-side quote in our pure order driven market.¹⁰

In our performance analysis we are interested in the performance of limit orders relative to market orders. For all four measures, we use the following definition:

$$\alpha_i = \delta_l^i - \delta_m^i \quad (5)$$

Based on the Handa and Schwartz (1996) hypothesis, we expect $\alpha_1 \geq 0$ for executed orders if the probability of the arrival of a liquidity trader is sufficiently large and $\alpha_1 \leq 0$ otherwise (adverse selection effect). We expect $\alpha_1 \leq 0$ for unexecuted orders (non-execution costs). Hence, when liquidity price changes can offset the adverse selection

¹⁰ In other markets, market orders can improve upon the existing opposite quote. The term “price improvement” is usually used to refer to this phenomenon [Knez and Ready (1996)].

cost, patient traders (who can afford to forego trading opportunities) prefer to trade via limit orders, and the relatively impatient traders prefer to trade via market orders.

Since $\delta_m^2 = 0$, the second relative performance measure becomes $\alpha_2 = \delta_l^2 - \delta_m^2 = \delta_l^2$.

This measure is always positive ($\alpha_2 > 0$) for executed limit orders, and is generally negative for unexecuted limit orders. One drawback of this second performance measure is its inability to account for the adverse selection cost since it is always positive for executed orders. To capture this cost, Harris and Hasbrouck (1966) construct another measure for executed orders that can capture such events. It is defined as:

$$\delta_i^3 = \begin{cases} B_5 - P_e & \text{for a buy order} \\ P_e - A_5 & \text{for a sell order} \end{cases} \quad i = m, l \quad (6)$$

where B_5 and A_5 are the bid and ask quotes prevailing five minutes after the execution, respectively. This ex post measure, which uses the same side quote as the reference, underestimates the information content of market orders and overestimates the adverse selection cost. Similarly, the measure that uses the other quote as a reference is likely to do the opposite. Therefore, in our analysis we also employ a fourth relative performance measure that uses the quote midpoint as a reference. This fourth measure is defined as:

$$\delta_i^4 = \begin{cases} Q_{15} - p_e & \text{for a buy order} \\ p_e - Q_{15} & \text{for a sell order} \end{cases} \quad i = m, l \quad (7)$$

where Q_{15} is the quote midpoint based on quotes prevailing fifteen minutes after the execution. The longer post-execution period used herein may be more appropriate for our less active market. By construction, $\alpha_4 < 0$ (and $\alpha_3 < 0$) reflects the adverse selection and

free option costs associated with limit orders. Positive values for a market order, $\alpha_4 > 0$ (and $\alpha_3 > 0$), can be interpreted as the information impact of the order.

4.2 The Empirical Findings for Relative Performance

We pool the data for all 56 stocks in the sample for the 65 trading days, and compute the daily mean for each measure, which forms the basic observation. During the sample period, the market rose with only 28 days with negative returns. This makes the distribution of returns for the full sample negatively skewed towards positive returns. As a result, the performance of sell orders is artificially higher. When the market is rising, sell limit orders are more likely to execute than buy limit orders placed equally far from the market. To deal with the same problem for their data set, Harris and Hasbrouck (1996) base their analysis on return-matched subsamples.¹¹ Applying the same approach to our sample produces results similar to those obtained using all orders in the sample. This can be attributed to the large difference in the subsample returns. In 22 of the 28 matched subsamples, the magnitude of the positive return is larger, which implies that the returns in these subsamples are still skewed towards positive returns. In our subsequent analysis we compute the performance measures using all orders in the sample.

Panel A in Table 4 reports the average values for the first relative performance measure (α_1) for sell and buy orders. The reported profitability of short selling by using a limit order is hypothetical since short selling is not a feasible strategy in this market. Consistent with Handa and Schwartz (1996), the relative performance is positive for executed limit orders and negative for nonexecuted limit orders. The relative performance

¹¹ They match all days with negative open-to-close S&P returns to the nearest positive return days, and exclude the days with largest positive returns.

of executed limit orders ranges from 0.3347 to 0.727 percent for all buy and sell order strategies and is significantly greater than zero. As expected, the performance is lower for more aggressive executed orders. The results indicate that information events do not dominate the gain from liquidity events in this market. Conversely, the relative performance of nonexecuted limit orders ranges from -0.0896 to -0.3811 for all buy and sell order strategies. The nonexecution cost is larger and significantly greater than zero for an at-the-quote limit order strategy. This nonexecution cost plays the role of classifying traders in this market into limit and market order traders depending on their relative patience. Overall, the limit order strategies (especially at or within the market) perform better than the market order strategy. The relative performance of all limit orders ranges from -0.024 to 0.2304 percent for all buy and sell order strategies and is significantly greater than zero for at-the-market for both sides, and for within-the-market for sell orders. This finding is not consistent with the strict prediction of Handa and Schwartz (1996) that a limit order must underperform a market order when execution is forced.

Please place Table 4 about here.

Panel B in Table 4 reports the averages of the second relative performance measure (α_2). While executed orders should have positive relative performances by definition, we observe that the relative performance based on this second performance measure for executed orders placed away from the market is the largest (as was the case for the first performance measure). The relative performance (α_2) is 3.8585 and 4.0027 for buy and sell orders placed away from the market, respectively. Because this performance measure severely penalizes limit orders placed away from the market which

do not execute, nonexecuted limit orders have negative performance. As anticipated, the most aggressive limit orders that do not execute have the lowest relative performance. The relative performance (α_2) is -0.705 and 2.9001 for nonexecuted buy and sell orders placed within the market, respectively. Similar to the findings for the first performance measure, the limit order strategy (especially at or within the market) performs better than the market order strategy. All the values for the second relative performance measure (α_2) are significantly greater than zero.

Panel C in Table 4 presents the average values of the two ex post relative performance measures (α_3 and α_4) for executed orders only. While both α_3 and α_4 are positive, this does not mean that the performance of executed limit orders per se is positive. In unreported results, we find that the average ex post performances of executed limit orders (δ_l^3) and market orders (δ_m^3) are -0.4357 and -1.6947 , respectively. This observation is puzzling because it suggests that market orders have no information content, which contradicts the finding that limit orders for the third performance measure are subject to adverse selection. On the other hand, the unreported averages of δ_l^4 and δ_m^4 are 0.55425 and -0.65925 , respectively. Since market orders have negative performances that are less than the tick size, the statistics suggest that market orders have no immediate impact on prices, and that executed limit orders benefit more from liquidity events. Accounting for the failure of this ex post measure to include nonexecuted orders, the overall results for the ex post relative measure further supports the finding that more patient traders will find limit order strategies more profitable.

5. Conclusions

In this paper we empirically analyze trading by limit versus market orders on the Saudi Stock Market. Our data set enables us to identify market and limit orders and a set of variables related to the state of the book and the order. Given the literature on a trader's choice problem in order driven markets, we interpret the data using a *Random Utility Model* and approach the problem statistically using a logit model. We also examine how the orders resulting from a trader's decision perform in the market.

Consistent with order driven market models, we find a negative spread effect on the probability of placing a market order. When the order imbalance increases in favor of the other side of the market, traders are more likely to submit market orders. When traders have optimistic expectations, the probability of placing a market buy order increases with short-term volatility. As the ratio of trade initiated by the same side of the market increases, the likelihood of the next order being a market order also increases. More active traders and traders with small orders are more likely to place market orders.

Findings for order performance show that executed limit orders have positive relative performance, while nonexecuted limit orders have negative relative performance. These findings suggest that patient traders prefer to trade via limit orders, while impatient traders prefer to trade via market orders. Overall, limit order strategies perform better than market order strategies.

References

- Ahn H., K. Bae and K. Chan, 2000, *Limit Orders, Depth, and Volatility*, forthcoming in Journal of Finance.
- Al-Suhaibani, M. and L. Kryzanowski, 2000a, *An Exploratory Analysis of the Order Book, and Order Flow and Execution on the Saudi Stock Market*, forthcoming in Journal of Banking and Finance.
- Al-Suhaibani, M. and L. Kryzanowski, 2000b, *The Information Content of Orders on the Saudi Stock Market*, forthcoming in Journal of Financial Research.
- Angel, J., 1995, *Limit versus Market Orders*, working paper, Georgetown University, School of Business Administration.
- Biais, B., P. Hillion and C. Spatt, 1995, *An Empirical Analysis of the Limit Order Book and the Order Flow in the Paris Bourse*, Journal of Finance 50, 1655-1690.
- Cohen, K., S. Maier, R. Schwartz and D. Whitcomb, 1981, *Transaction Costs, Order Placement Strategy, and Existence of the Bid-ask Spread*, Journal of Political Economy 89, 287-305.
- Foucault, T., 1999, *Order Flow Composition and Trading Costs in a Dynamic Limit Order Market*, Journal of Financial Markets, 2, 99-134.
- Foucault, T., A. Röell and P. Sandås, 1999, *Imperfect Market Monitoring and SOES Trading*, working paper.
- Greene, W., 1997, *Econometric Analysis*, Prentice Hall.

- Handa, P. and R. Schwartz, 1996, *Limit Order Trading*, Journal of Finance, 51, 1835-1861.
- Handa, P., R. Schwartz and A. Tiwari, 1997, *The Determinants of Bid-ask Spread in an Order-driven Market*, working paper.
- Harris, L., 1998, *Optimal Dynamic Order Submission Strategies in some Stylized Trading Problems*, Financial Markets, Institutions & Instruments, 7:2.
- Harris, L. and J. Hasbrouck, 1996, *Market vs. Limit Orders: The SuperDOT Evidence on Order Submission Strategy*, Journal of Financial and Quantitative Analysis, 31:2, 213-231.
- Hollifield, B., R. Miller and P. Sandas, 1996, *An Empirical Analysis of a Pure Limit Order Market*, working paper, Carnegie Mellon University.
- Knez, P and M. Ready, 1996, *Estimating the Profit from Trading Strategies*, The Review of Financial Studies 9, 1121-1163.
- Lo, A., C. MacKinlay and J. Zhang, 1998, *Econometric Models of Limit Order Executions*, NBER Working Paper No. W6257.
- Maddala, G., 1983, *Limited-dependent and Qualitative Variables in Econometrics*, Cambridge University Press.
- Parlour, C., 1998, *Price Dynamics in Limit Order Markets*, The Review of Financial Studies 11:4, 789-816.
- Saudi Arabian Monetary Agency, 1992, *Electronic Securities Information System: Instructions to Central Trading Units*.

Table 1**Summary Statistics for Market and Limit Orders on the Saudi Stock Market**

The percentage of classified market and limit orders are reported after pooling all stocks for all 65 trading days in the period between October 31, 1996 and January 14, 1997. The orders are classified by order direction (sell, buy), size (number of shares), whether the order is part of a package or not, and by the number of orders in the package that generated the order. An order package contains at least two orders. The total number of orders is 267,518.

Order Characteristics	Order Type			All
	Market	Limit		
		Executed	Unexecuted	
	%	%	%	%
Side				
Buy	55.41	45.46	46.97	48.88
Sell	44.59	54.54	53.03	51.12
Size				
≤ 100	27.28	33.94	11.72	23.82
101-500	41.42	37.79	35.14	37.86
501-1000	19.62	17.72	26.48	21.50
> 1000	11.68	10.55	26.66	16.82
Part of a package				
Yes	86.46	83.63	84.44	84.74
No	13.54	16.37	15.56	15.26
Orders per package				
2 - 5	33.45	43.29	48.79	35.95
6 - 10	17.01	18.10	17.99	15.03
11 - 15	10.65	10.39	09.84	08.70
> 15	38.89	28.22	23.38	25.06
All orders	28.76	34.32	36.91	100.00

Table 2**Summary Statistics for the Explanatory Variables**

The table reports the summary statistics for the explanatory variables in the logit regressions. The *inside spread* is the difference between the first best ask and the first best bid. *Order imbalance* is the depth at the same side divided by total depth. *Volatility* is the ratio of trades (market orders) in the last half-hour of trading to trades in the last one hour of trading. *Prior market order* is the ratio of the market orders that are initiated by the same side of the market to the total market orders submitted during the last half-hour. *Size* is the number of shares in the order divided by 100. *Orders per package* is the number of orders in the package that generated the order. The sample used includes only orders preceded by a valid bid-ask spread.

<i>Variables</i>	<i>Buy Orders</i>		<i>Sell Orders</i>	
	<i>Mean</i>	<i>Standard deviation</i>	<i>Mean</i>	<i>Standard deviation</i>
<i>Inside Spread</i>	1.914	2.05	1.856	1.798
<i>Order imbalance</i>	0.522	0.332	0.503	0.33
<i>Volatility</i>	0.593	0.304	0.588	0.305
<i>Prior market order</i>	0.54	0.336	0.452	0.336
<i>Order size</i>	8.44	17.341	8.091	15.569
<i>Orders per package</i>	19.887	29.546	9.752	18.635

Table 3**Logit Regression Results**

This table reports the results for the logit regression, $E[y|x] = \Lambda(x'\beta)$, using the pooled data for the 56 stocks in the sample. y is a dummy variable that equals one if the order is market, and zero otherwise. The set of regressors in x includes a constant, inside spread, order imbalance, order size, number of orders per package, prior market order, and volatility as they are defined in Table 2. $\Lambda(\cdot)$ is the logistic cumulative distribution function, $e^{x\beta} / (1 + e^{x\beta})$. The coefficient is the β estimate. The slope is the marginal effect of x on the probability, which is equal to $\Lambda(x'\beta)[1 - \Lambda(x'\beta)]\beta$ when $\Lambda(x'\beta)$ is evaluated at the mean of the regressors. The McFadden pseudo R-squared is $1 - (\ln L / \ln L_0)$, where $\ln L$ and $\ln L_0$ are the log-likelihood functions evaluated at the unrestricted and restricted estimates (all coefficients, except the constant, are zero), respectively.

	Buy Orders		Sell Orders	
<i>No. of observations</i>	114083		111608	
<i>Dependent variable equals one (%)</i>	33.51		26.81	
<i>Proportion of orders that are correctly classified (%)</i>	69.7		74.03	
<i>McFadden pseudo R-squared</i>	0.0973		0.0895	
<i>Independent Variables</i>	<i>Coefficient</i>	<i>Slope</i>	<i>Coefficient</i>	<i>Slope</i>
<i>Constant</i>	-1.7618*		-1.8958*	
<i>Inside spread</i>	-0.1526*	-0.0325*	-0.0846*	-0.0155*
<i>Order imbalance</i>	0.4968*	0.106*	0.731*	0.1336*
<i>Volatility</i>	0.2104*	0.0449*	-0.1153*	-0.0211*
<i>Prior market order</i>	1.749*	0.373*	1.7145*	0.3134*
<i>Order size</i>	-0.0343*	-0.0073*	-0.0383*	-0.007*
<i>Orders per package</i>	0.0104*	0.0022*	0.0142*	0.0026*

* Statistically significant at the 1% level. The hypothesis that all the coefficients, except the constant, are zero is rejected for all stocks using both the Likelihood ratio and Wald tests.

Table 4
Average Relative Order Performance

This table reports the average relative performance of limit buy and sell orders measured as $\alpha_i = \delta_l^i - \delta_m^i$, where δ_l^i is the performance of a limit order (l) and δ_m^i is the performance of a corresponding market order (m) for relative performance measure i . The relative performance reported in each cell is the simple average of the observations on the 65 daily means. The means are classified by aggressiveness. *Away* is the limit order placed away from the same-side best quote. *At* is the limit order placed at the best same-side quote. *Within* is the limit order placed within the prevailing inside spread. The Wilcoxon non-parametric test is used to test the null hypothesis that the median return (of the 65 daily means) equals zero versus the alternate hypothesis that the median return is greater than zero.

	Buy Orders			Sell Orders		
Panel A: First Relative Performance Measure (α_1) %						
	Away	At	Within	Away	At	Within
Executed limit orders	0.4526**	0.3544**	0.3347**	0.4727**	0.3684**	0.3424**
Nonexecuted limit orders	-0.0896	-0.2118*	-0.1984	-0.1132	-0.3811**	-0.2541
All limit orders	-0.024	0.1882**	0.2268	-0.0385	0.1645**	0.2304**
Panel B: Second Relative Performance Measure (α_2)						
	Away	At	Within	Away	At	Within
Executed limit orders	4.0027**	1.5624**	2.2399**	3.8585**	1.5095**	1.6467**
Nonexecuted limit orders	-1.4451**	-1.335**	-2.9001**	0.4129**	-0.4024**	-0.705**
All limit orders	-0.9235**	0.6372**	0.865**	0.9638**	1.0596**	1.2316**
Panel C: Third and Fourth Relative Performance Measures (α_3, α_4)						
	Away	At	Within	Away	At	Within
α_3	1.716**	1.2869**	0.8119**	1.8835**	1.2048**	0.6408**
α_4	2.2762**	1.0599**	0.8195**	2.306**	1.1001**	1.1499**

** Indicates significance at the 1% level.

* Indicates significance at the 5% level.

Figure 1
Probability of Placing a Market Order

This figure plots the probabilities of placing a market sell and a market buy order using the regression model explained in table 3. The probability is estimated using: $1/(1+e^{-x\beta})$. Each subplot shows the probability plotted over the range of the specified explanatory variable, while keeping the other explanatory variables constant at their sample means. The dashed and solid lines represent buy and sell orders, respectively.

